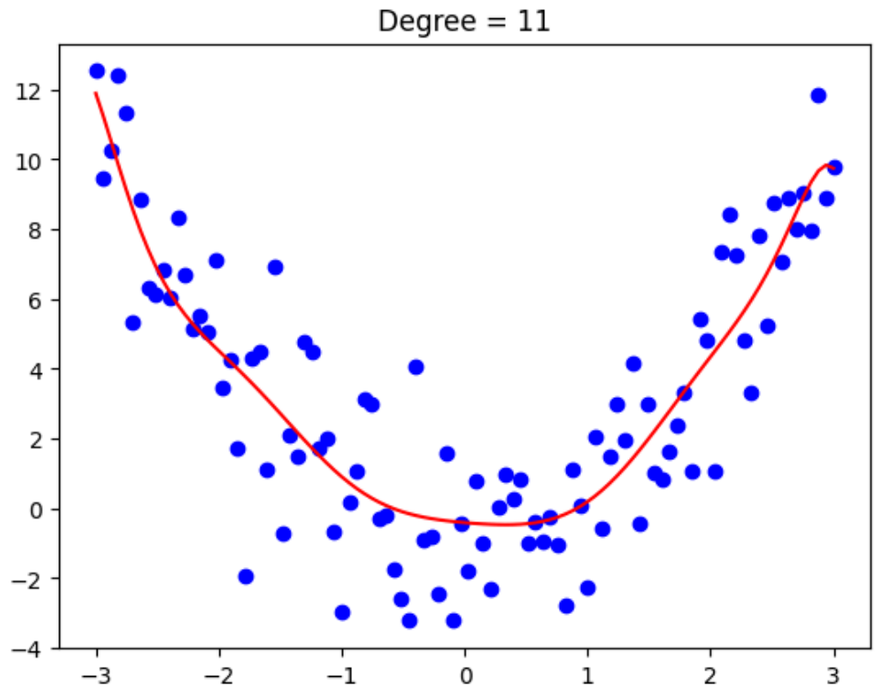
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split

# 🔷 Step 1: Dummy data
np.random.seed(0)
X = np.linspace(-3, 3, 100).reshape(-1, 1)
y = X**2 + np.random.randn(100, 1) * 2   # quadratic data + noise

# 🔷 Step 2: Train-Test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# 🔷 Step 3: Polynomial Features (try different degrees)
degree = 11   # change this to 1, 2, 10 to see behavior
poly = PolynomialFeatures(degree=degree)

X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

# 🔷 Step 4: Train model
model = LinearRegression()
model.fit(X_train_poly, y_train)

# 🔷 Step 5: Predictions
y_train_pred = model.predict(X_train_poly)
y_test_pred = model.predict(X_test_poly)

# 🔷 Step 6: Errors
train_error = mean_squared_error(y_train, y_train_pred)
test_error = mean_squared_error(y_test, y_test_pred)

print("Training Error:", train_error)
print("Test Error:", test_error)
```

```
Training Error: 3.5629671378238057
Test Error: 2.793262553002291
```

```
plt.scatter(X, y, color='blue')
X_range = np.linspace(-3, 3, 100).reshape(-1, 1)
X_range_poly = poly.transform(X_range)
y_range_pred = model.predict(X_range_poly)

plt.plot(X_range, y_range_pred, color='red')
plt.title(f"Degree = {degree}")
plt.show()
```